

On code design and bit operations for mash distance

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Bit operations for mash distance

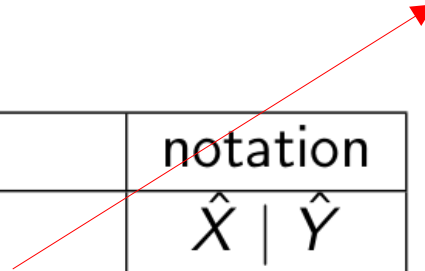
Estimation of Jaccard index :

$$J(A, B) \approx \frac{|S(A \cup B) \cap S(A) \cap S(B)|}{|S(A \cup B)|}$$

and $S(A \cup B) = S(S(A) \cup S(B))$

sketch operation	column operation	notation
$X \cup Y$	bitwise or	$\hat{X} \mid \hat{Y}$
$S(X)$	keep the first s ones of $\hat{X}$	$\hat{X}_s$
$S(X) \cap S(Y)$	bitwise and	$\hat{X} \& \hat{Y}$
$ X $	sum along $\hat{X}$	$\sum \hat{X}$

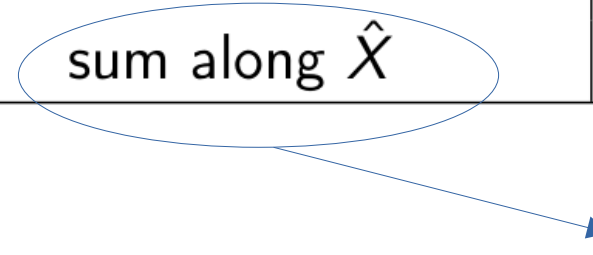
This is slower
than the rest



Total operation :

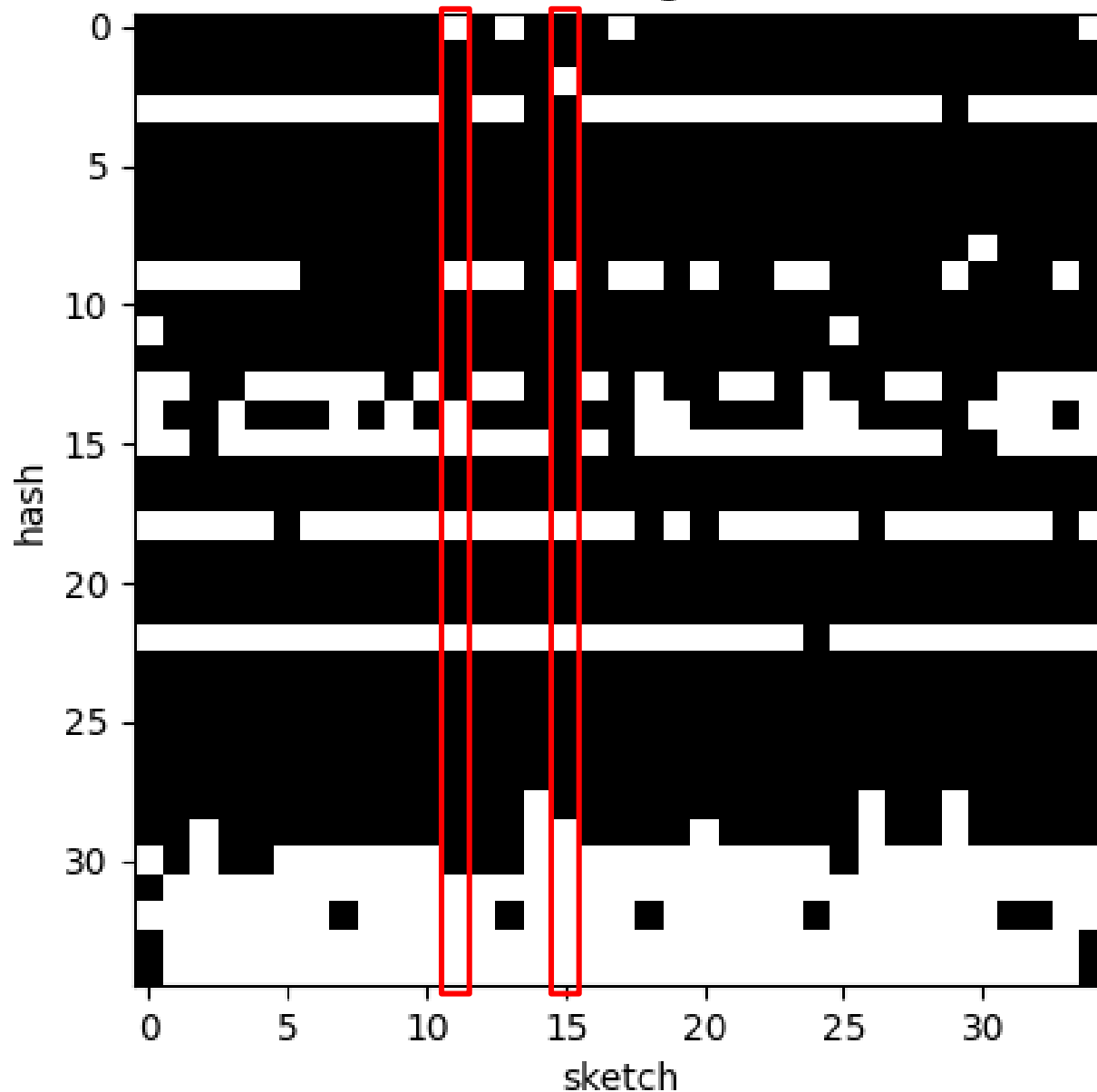
$$\frac{\sum \left((\hat{A}_s \mid \hat{B}_s)_s \& \hat{A}_s \& \hat{B}_s \right)}{s}$$

popcount or rank_1



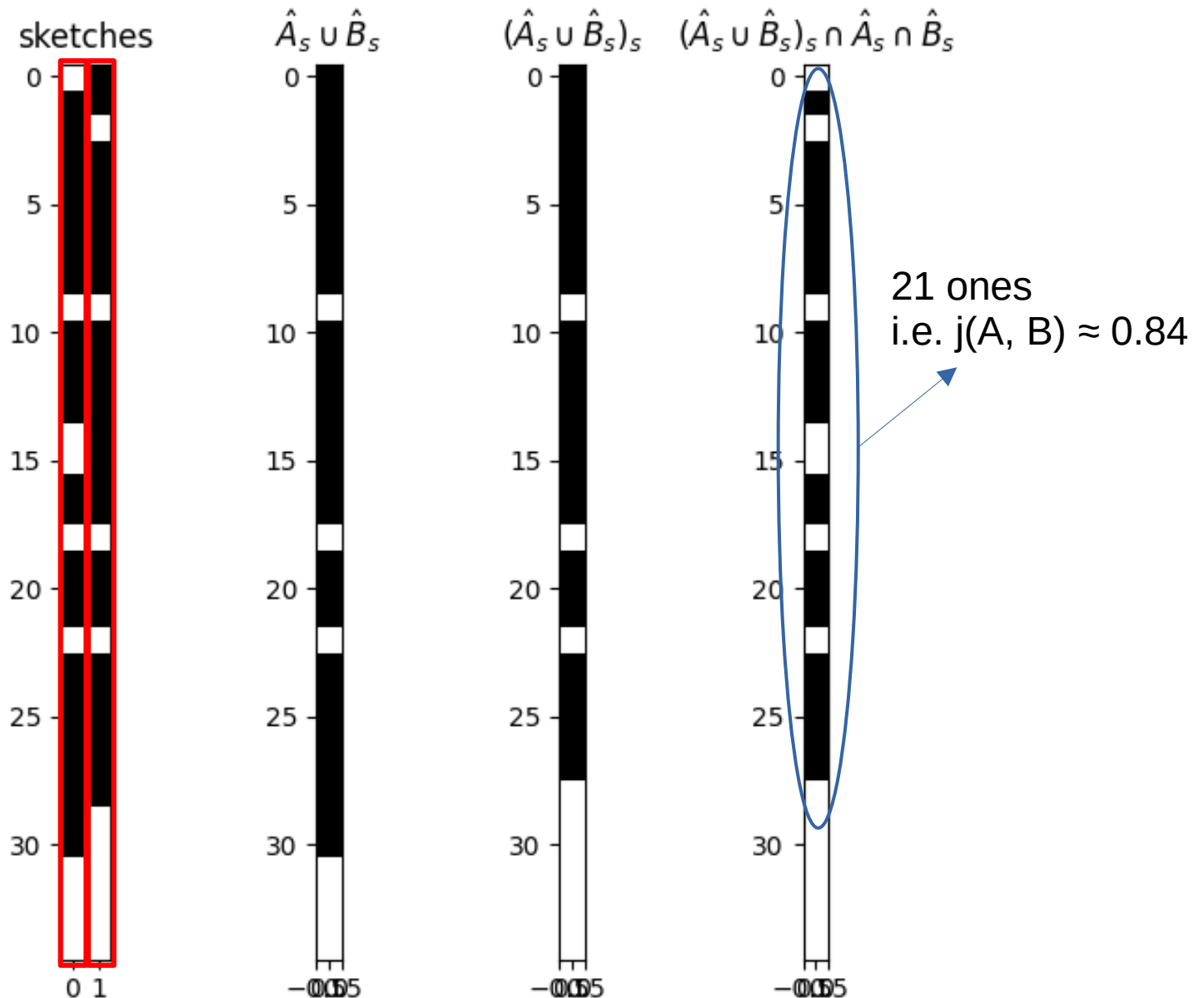
Bit operations for mash distance

Presence-absence matrix for ngonono - 35 sketches - $s = 25$



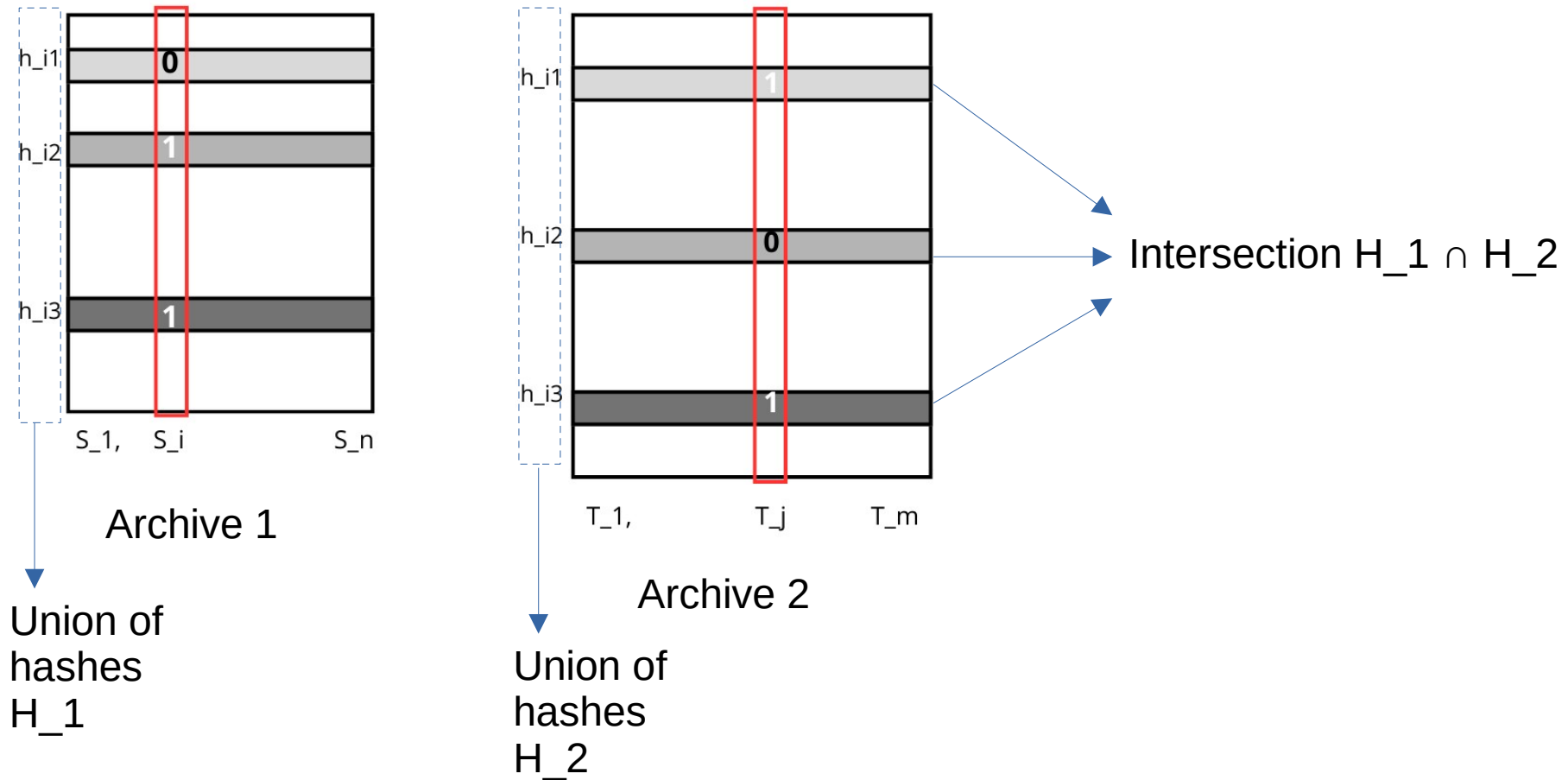
$s = 25$

Bit operations for mash distance

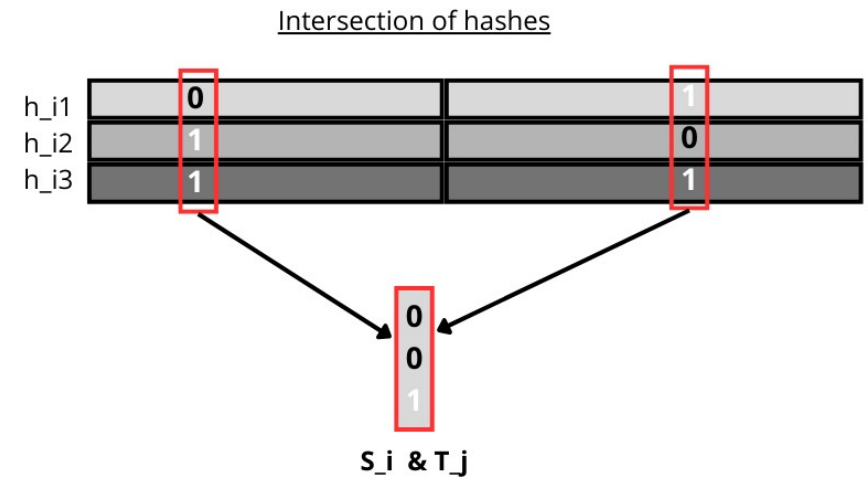
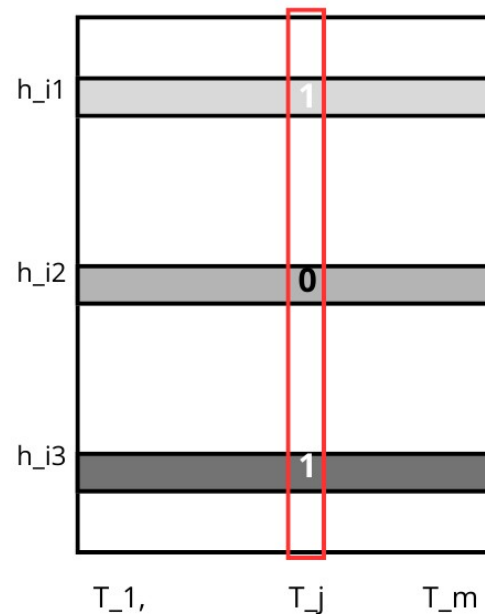
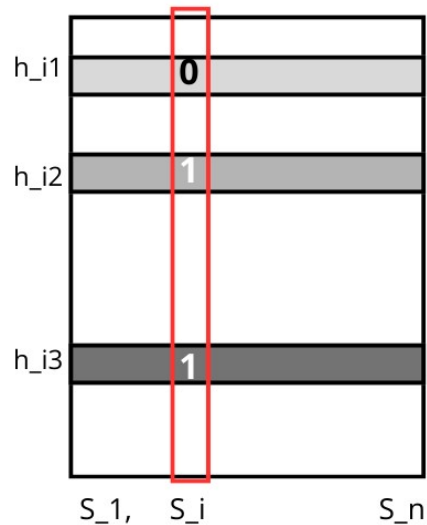


$s = 25$

Mash distance between different archives



Mash distance between different archives



$$\frac{\sum \left((\hat{A}_s \mid \hat{B}_s)_s \& \hat{A}_s \& \hat{B}_s \right)}{S}$$

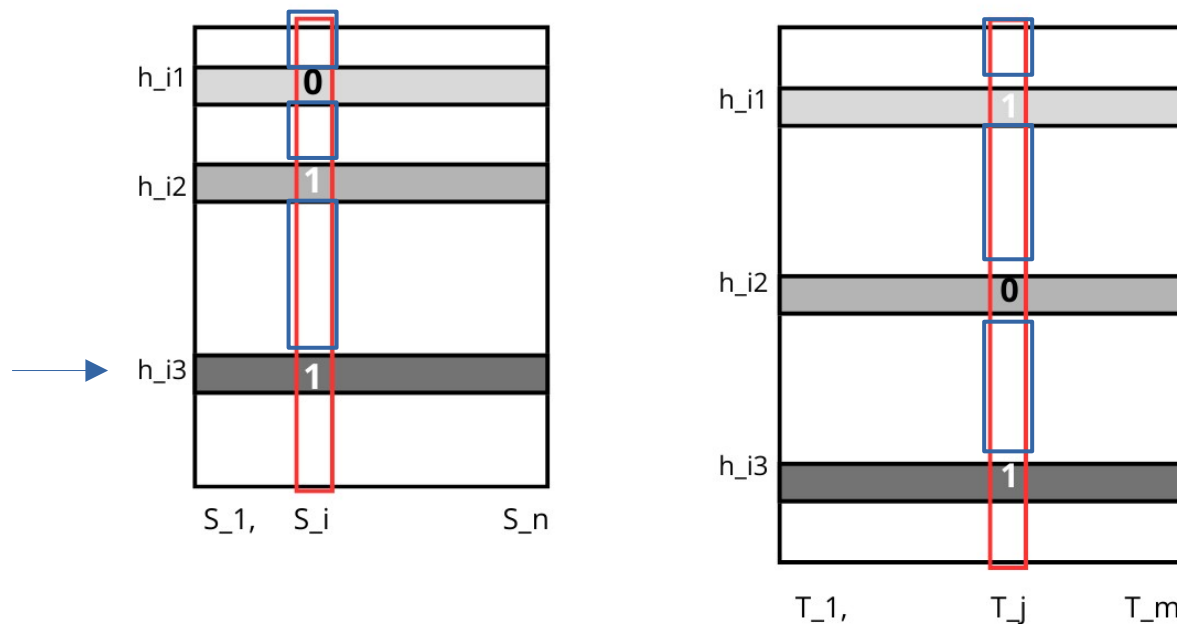
ISSUE : ~~how do we compute the sketch of the union ?~~

how do we get the number of ones before each h_{ij} ?

Mash distance between different archives

ISSUE : ~~how do we compute the sketch of the union ?~~

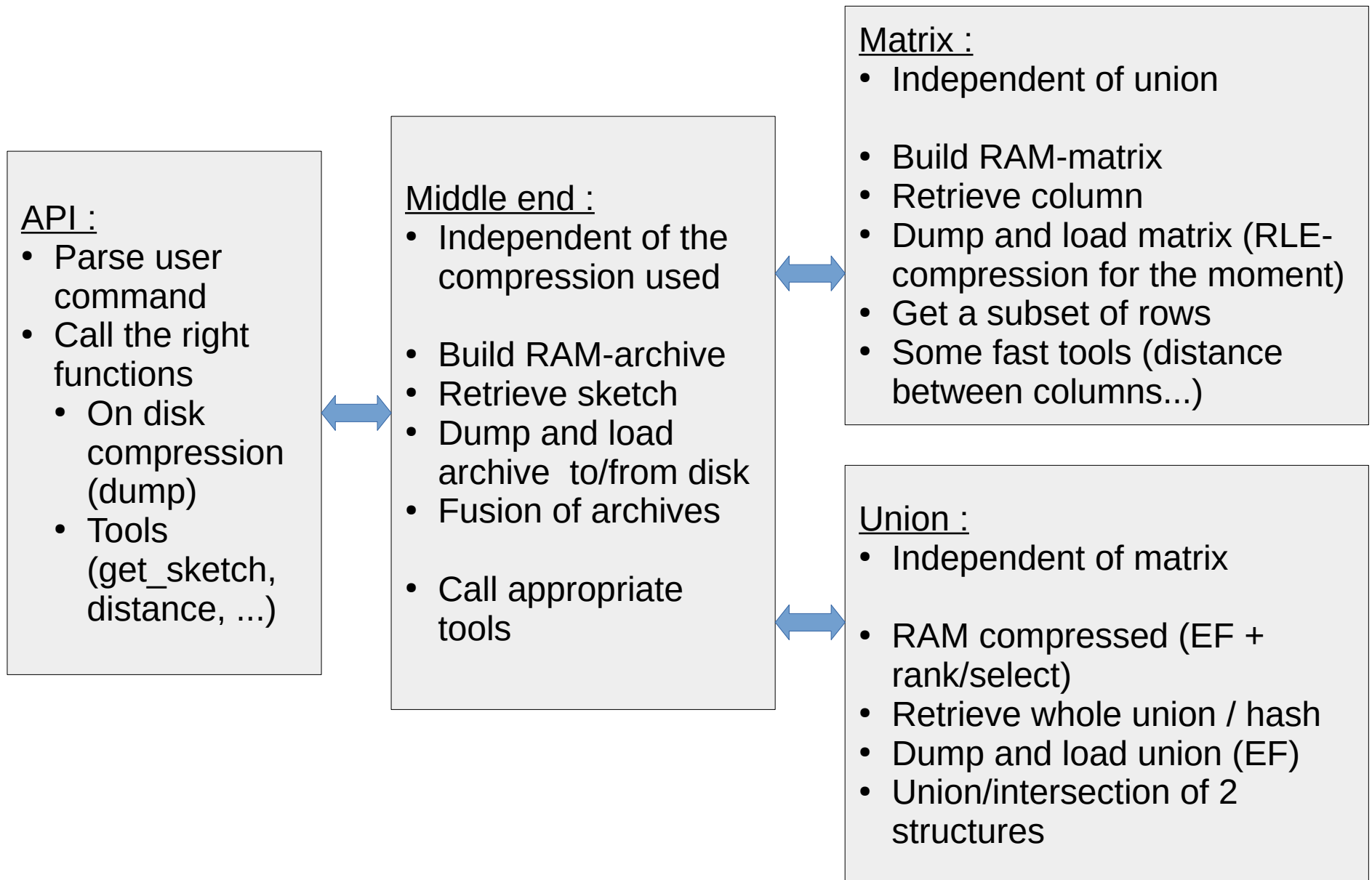
how do we get the number of ones before each h_{ij} ?



Nb ones in + nb ones in intersection before h_{ik}

➤ rank and select in matrix ?

Code design



Pros and cons of code design

Pros :

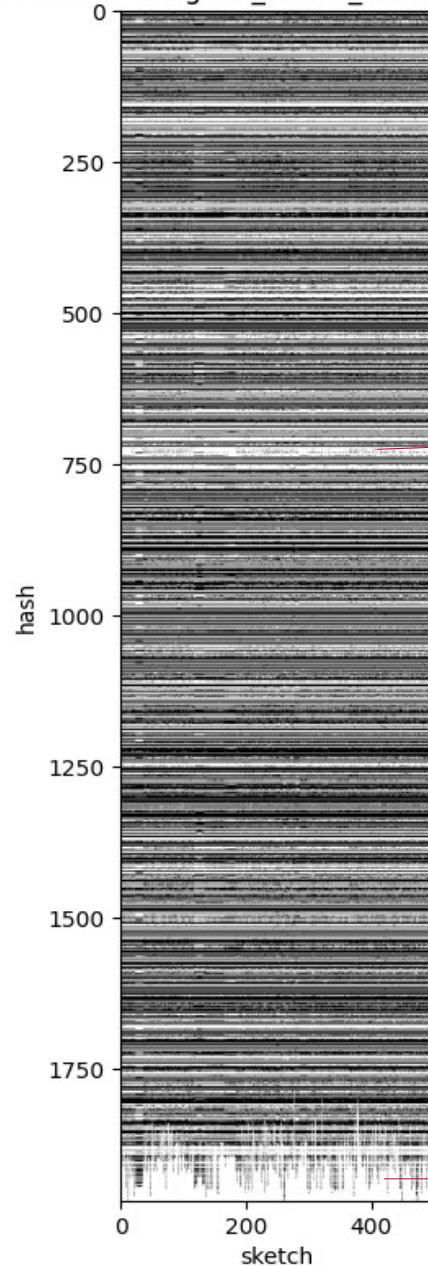
- Easy to change compression techniques (heritage + encapsulation)
- Default RAM compression
- Easy to add new elements in compression pipeline

Cons :

- No disk compression without RAM-compression beforehand (add tool ?)
- Compression is not best fitted for use (e.g. EF-compression does not always require rank/select)

Quick recap of past results

Presence-absence matrix for ngono_s1000_S500 - 500 sketches - s = 1000

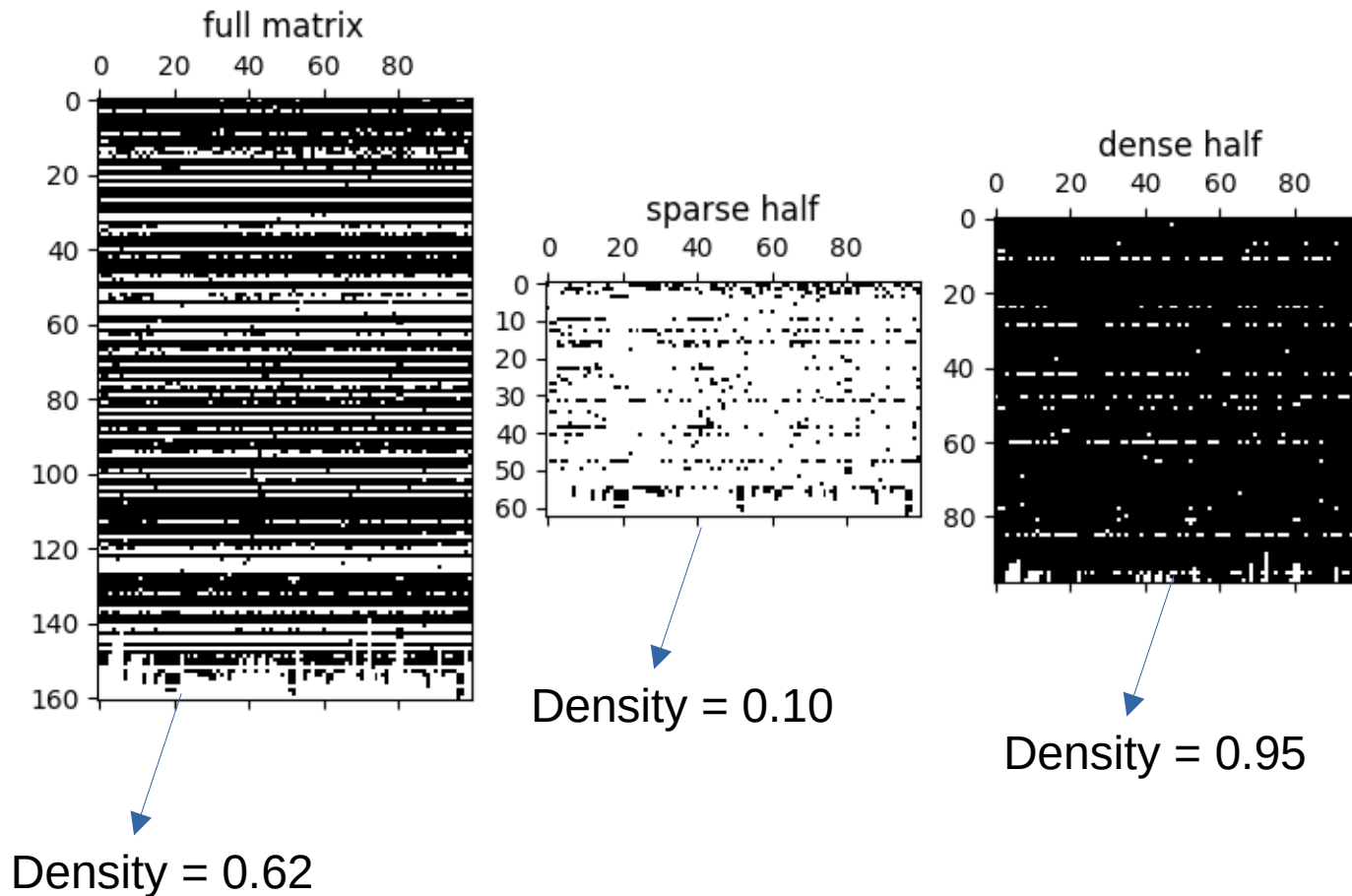


► Sparsity comes from sparse rows

► Small sparse area at the end

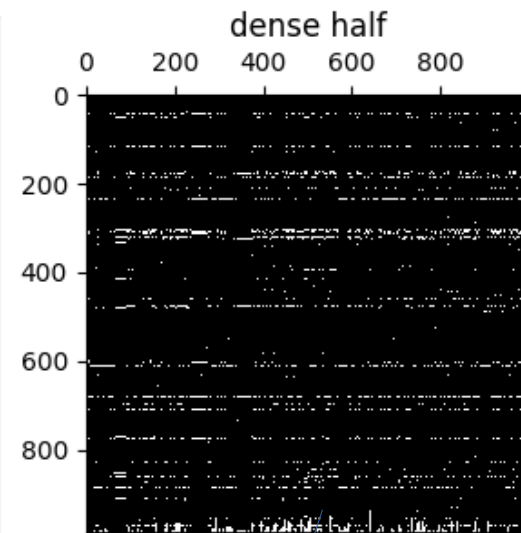
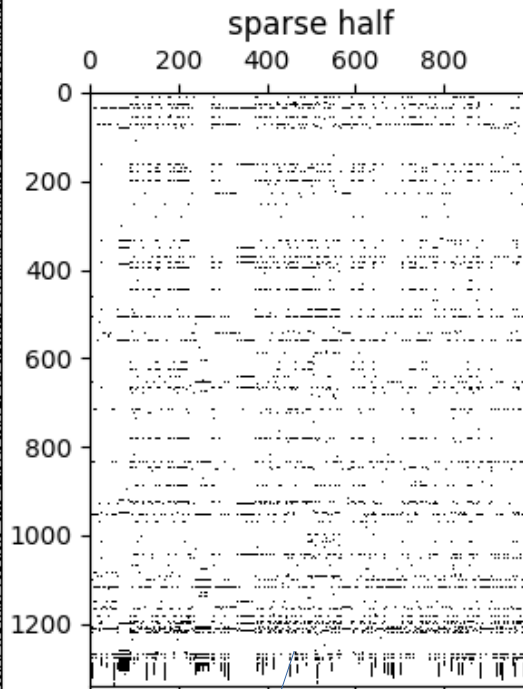
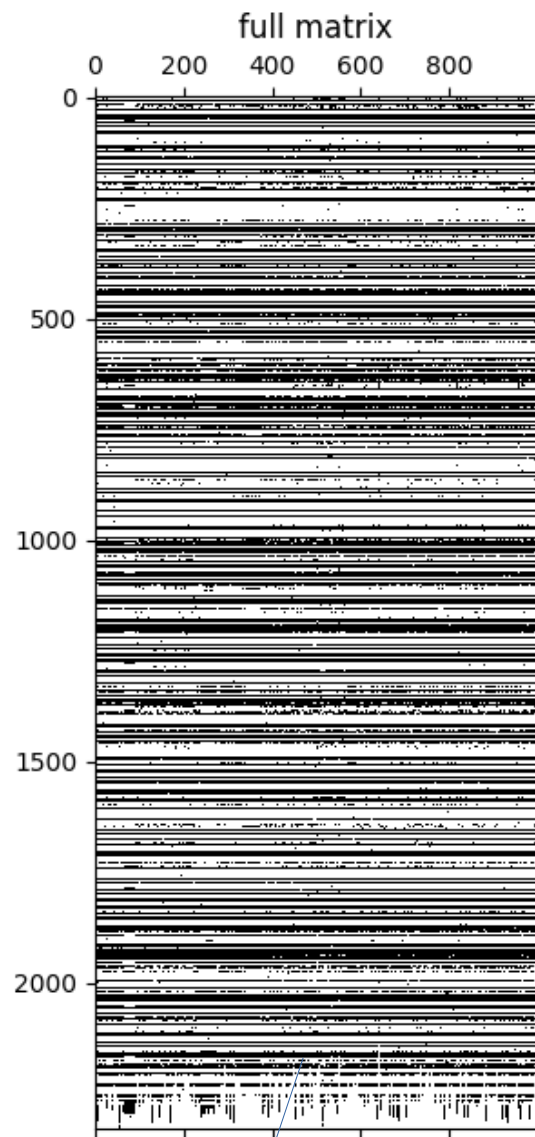
- $g = 500$
- $s = 1000$
- Density of matrix : 51%
- Delta = 29.754

Splitting the matrix : first tries



$s = 100$
 $g = 100$

Splitting the matrix : first tries



Density = 0.04

Density = 0.96

Density = 0.43

$s = 1000$
 $g = 1000$

Conclusion

Conclusions :

- Splitting the matrix allows for very dense and sparse parts
- Computing mash distance on 2 sketches from the same archive is very easy and fast
- It is trickier if the archives are different

To do :

- Benchmark density and compressibility for split matrices according to s and g
- Work on implementation
- Benchmark compressibility of not-split matrix